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(51) International classification :H02S10/12, G05F1/67, F03D9/25, H02S20/25, F03D9/45 (86) International Application No :NA Filing Date :NA (87) International Publication No : NA (61) Patent of Addition to Application Number :NA Filing Date :NA (62) Divisional to Application Number :NA Filing Date :NA		(71)Name of Applicant : 1)Chandigarh University Address of Applicant :Chandigarh University, National Highway 05, Chandigarh-Ludhiana Highway, Mohali, Punjab -140413, India Mohali ----- -- ----- Name of Applicant : NA Address of Applicant : NA (72)Name of Inventor : 1)Dr. Shivani Singla Address of Applicant :Department of Physics, Chandigarh University, National Highway 05, Chandigarh-Ludhiana Highway, Mohali, Punjab -140413, India Mohali ----- ----- 2)Heemal Jaglan Address of Applicant :Department of Computer Science and Engineering, Chandigarh University, National Highway 05, Chandigarh-Ludhiana Highway, Mohali, Punjab -140413, India Mohali ----- ----- 3)Anushka Shekhawat Address of Applicant :Department of Computer Science and Engineering, Chandigarh University, National Highway 05, Chandigarh-Ludhiana Highway, Mohali, Punjab -140413, India Mohali ----- ----- 4)Naveen Bansal Address of Applicant :Department of Computer Applications, University Institute of Sciences, Chandigarh University, National Highway 05, Chandigarh-Ludhiana Highway, Mohali, Punjab -140413, India Mohali ----- ----- 5)Dr. Nupur Aggarwal Address of Applicant :Department of Physics, Chandigarh University, National Highway 05, Chandigarh-Ludhiana Highway, Mohali, Punjab -140413, India Mohali ----- -----
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(57) Abstract :

The present invention relates to a rooftop tile system integrated with miniature vertical-axis wind turbines (VAWTs) for converting wind energy into electrical energy. Each tile comprises a compact VAWT coupled to a micro-generator and a smart micro-inverter with Maximum Power Point Tracking (MPPT) 22 for efficient energy conversion. Conductive tracks embedded within the tile route the generated electricity to a centralized controller, battery, or building grid. The system optionally includes a piezoelectric support layer for harvesting vibrational energy, and integrated photovoltaic strips for hybrid solar-wind operation. Additional features include aerodynamic wind-guiding vanes, a self-cleaning nano-coating, drainage channels, IoT-based real-time monitoring, and protective systems such as lightning arrestors and thermal sensors. The invention offers a modular, low-noise, and aesthetically integrated renewable energy solution for urban and residential buildings, enhancing rooftop space utilization while contributing to decentralized and sustainable power generation.

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